USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 21 LAB - Antenna Pattern
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

This packet is the L21 Lab, in three parts, and you turn it in once, complete, before Lesson 22. **Part 1, Predict**, is yours to do before the lab period: what each Part 2 table should read on your kit. **Part 2, Measure**, is your own kit's readings, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict Tables A to E.** Fill in each table below with what the matching Part 2 table should read on your kit, at the HB100 frequency your kit reported with **Find HB100** in the L18 Lab. If you did not record one, use the nominal 10.525 GHz and say so. Lesson 21's Pre-Lab frame names the frame that works each method, at 10.0 GHz.

   The array has $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch. Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. The HB100 stands 1.2 m from the array. A sweep by hand carries it from -90° to $+90^\circ$ along an arc of that radius in about 22 s, and your radius drifts by up to 10 cm either way. Report levels as Part 2 does: Tables B and C in dBc, and Tables D and E relative to the Table D reference, the level with the kit at 0° and the beam at 0°.

Setup	**Frequency**	**Far-field distance**	**Slowest update rate (/s)**
Table A: stand reads	$+30$	0	-30
Angle (Peak)			
Table B: beam sweep	**HPBW**	**Left SL (dBc)**	**Right SL (dBc)**
Table C: by hand	**Peak spread (dB)**	**Left SL (dBc)**	**Right SL (dBc)**

Table D, 0°	**Angle**	**Level**
Main lobe		
Left first null		
Right first null		
Left first SL		
Right first SL		

Table E, 30°	**Angle**	**Level**
Main lobe		
Inner first null		
Outer first null		
Inner first SL		
Outer first SL		

Show your work for one row of each table.

Part 2: Measure (your kit, in class)

Follow the procedure on the L21 Lab page. Before you record anything, set **Rx Gain** so that the main-lobe peak, with the source at boresight and the beam at 0°, reads -10 ± 3 dBFS on the FFT, and leave it there for the whole lab. Keep the HB100 at patch-row height, 1.2 m from the array. Record each reading as you take it.

Signal Freq (Find HB100) =		Rx Gain =	
Source distance =		Updates/s =	

2. **Table A: check the angle axis.** With the beam sweeping as in the L18 Lab, turn the kit so the source sits at each angle below, and record the **Angle** readout, set to **Peak**. Turning the kit's face toward E1, as seen from the source, puts the source on the $+$ side, and the stand's scale reads the opposite sign: -30 on the stand puts the source at $+30^\circ$.

Source angle (stand reads)	-30° ($+30$)	0° (0)	$+30^\circ$ (-30)
Angle (Peak)			

3. **Table B: the beam sweep.** Kit back at 0°, beam sweeping. Read the trace.

Peak (dBFS)	HPBW	Left sidelobe (dBc)	Right sidelobe (dBc)

4. **Table C: the azimuth sweep by hand.** Beam held at 0°. Carry the HB100 from -90° to $+90^\circ$ on the taped 1.2 m arc, aimed at the array, three times. Record each sweep's main-lobe peak and both first sidelobes.

	Sweep time (s)	Peak (dBFS)	Left SL (dBc)	Right SL (dBc)
Sweep 1				
Sweep 2				
Sweep 3				
Spread (max − min)				

5. **Table D: the pass on the stand, beam held at $0°$.** HB100 on its tripod at boresight, 1.2 m out, beam sweeping. One pass fills Tables D and E: each sweep sample is the beam held at that steer angle, so the trace at $0°$ steer is this table and the sample nearest $30°$ is Table E. Turn the kit to each mark of the stand and record the trace at $0°$ relative to its value with the kit at $0°$, in dB, at the source angle: the stand reading with its sign flipped, as in Table A. Write "floor" where the reading stops changing as you turn.

θ_{src}	$-90°$	$-80°$	$-70°$	$-60°$	$-50°$	$-40°$	$-30°$	$-20°$	$-10°$	$0°$
Level										

θ_{src}	$+10°$	$+20°$	$+30°$	$+40°$	$+50°$	$+60°$	$+70°$	$+80°$	$+90°$	
Level										

Reference: kit at $0°$, trace at $0°$ steer (dBFS) = ____

Now turn the kit slowly through each lobe and null below, and record the angle and level of its extreme, the highest reading for a lobe and the lowest for a null. Record every level relative to the reference above, as in the marks; Part 3 converts the lobes to dBc. Last, turn back to the main lobe and the right first sidelobe and read each again: the change is the stand's repeatability.

	Source angle	**Level**
Main lobe		
Left first null		
Right first null		
Left first sidelobe		
Right first sidelobe		
Main lobe, read again		
Right first sidelobe, read again		

6. **Table E: the pass on the stand, beam held at 30°.** The same pass, read at the sweep sample nearest 30°. Record every level, in the marks and in the lobe table, relative to the Table D reference.

θ_{src}	-90°	-80°	-70°	-60°	-50°	-40°	-30°	-20°	-10°	0°
Level										

θ_{src}	$+10^\circ$	$+20^\circ$	$+30^\circ$	$+40^\circ$	$+50^\circ$	$+60^\circ$	$+70^\circ$	$+80^\circ$	$+90^\circ$	
Level										

	Source angle	**Level**
Main lobe		
Inner first null		
Outer first null		
Inner first sidelobe		
Outer first sidelobe		

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions. Convert the lobe levels of Tables D and E to dBc first: subtract that table's main-lobe level.

7. **Broadside, three ways.** Set your beam sweep (Table B), your three sweeps by hand (Table C), and the pass on the stand at 0° (Table D) beside your predictions. Which features do the three methods agree on, and which do they not? Build the error budget that decides it from your own readings, and give a cause for each disagreement you find.

8. **The beam at 30°.** Compare the two first sidelobes in Table E with each other, with your predictions, and with your broadside pair in Table D. What do you conclude caused the difference you see, and how sure can you be from these readings? Describe a measurement you could add on this kit that would test your conclusion.

9. **By hand against the stand.** You measured the same held beam by carrying the source and by turning the kit. From the spread of your sweeps in Table C and your repeated readings in Table D, decide which quantities you would report from each method, and why. Then state this array's broadside first-sidelobe level as you would report it from today's data, with its uncertainty, and say what it doesn't explain.

Documentation: _____